Phase 3: Design Analysis Report

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1. Introduction

Farm Boy Lakeshore experiences significant difficulties in managing frequent price changes for grocery items. Price tags are often updated incorrectly due to errors in identifying weekly, monthly, and in-store promotions. The lack of an organization leads to pricing inconsistencies, customer dissatisfaction, and increased workload for employees who must manually verify and correct pricing errors.

The company is legally bound to honour the displayed price, leading to a cut in revenue if the price is lower than it should be. Loyal shoppers grow frustrated when a price is higher than it was advertised, especially given employees can not help as they do not know the intended price themselves.

Problems also arise from having an ununified system. Immediate price changes for expiring items, surpluses, etc are done in-store by department managers without notifying the head office of the changes. The head office is often dissatisfied with their changes creating a need for a reviewing process.

To address these challenges, we propose a streamlined digital solution that integrates seamlessly into the store’s existing Google-based web applications. This system will automate price tracking, improve accuracy, and enhance operational efficiency, ensuring that price changes are correctly implemented and monitored.

2. Requirements Analysis

The proposed system must meet the following requirements:

- Employees need a way to accurately track price changes efficiently.
- Distinguish between weekly, monthly, and in-store price changes to prevent confusion.
- Include a category for in-store price changes with a request and review system in order to mitigate confusion between the store's floor and head office.
- A system that organizes price tag updates by aisle and date for easy retrieval
- Users must be able to receive printable versions of corrected pricing information quickly.
- Employees need a system that works with existing workflows and doesn’t require installing additional software.
- Ensure usability for employees with minimal training requirements.
- Potentially integrate with the head office system for real-time pricing updates.
- Employees and managers need a way to ensure only authorized personnel can approve significant price changes.
- Provide an audit log feature to track all pricing changes, ensuring accountability and transparency in the process.
- Farm Boy uses physical tags and signs for all pricing in the store. Any item going on sale has its default tag replaced with a bright special tag to catch the eye. When they go back to regular pricing, these tags are removed. Before new sale prices activate on Thursday, it is crucial that the new tags are delivered to the store on Wednesday morning otherwise prices will mislead customers and limit sales. When mistakes occur, having a system for promptly communicating with the head office is essential for swift resolution.

3. Proposed Solution
We propose developing a web-based application that will function within the store's already existing Google-based employee portal. The key features of this solution include:

- Automated Price Tracking: The system will maintain records of price changes, sorted by aisle and date, allowing employees to quickly identify necessary updates.

- Change Identification: The system will highlight outdated price tags and indicate whether changes are weekly, monthly, or in-store promotions.

- Filtering and Searchability: Employees will be able to filter price updates by department, aisle, and date to quickly access relevant information.

- Printable Reports: A feature to generate and print lists of necessary price updates for weekly (also monthly and exclusive in store prices when applicable) store operations. Two lists will be printed, one will consist of the expiring sales to remove while the other will be the new price changes to put up. This system is not exclusive to Wednesday, the unified web portal will display a notification upon opening the page that the appropriate department has expiring price changes that need to be done soon to account for in store exclusive prices.

- Potential Head Office Integration: The system will sync with the head office pricing system to receive real-time pricing updates, reducing errors caused by incorrect price tags from management. Assuming the head office system offers API access or data export functionality, we will use real-time syncing libraries (e.g., Firebase or RESTful APIs) to receive updates. This integration is contingent on cooperation from IT at head office. The system will undergo front-end and back-end security testing following OWASP guidelines to ensure data integrity and prevent unauthorized access.

- Employee Login through Existing Portal: Since all store operations are managed through Google web apps, this system will integrate directly into the portal, ensuring a smooth and familiar user experience.

- Price change authorization: Managers in store will no longer have unrestricted control over pricing. If a price falls below the cost price or is significantly above the standard pricing, the system will require authentication from the head office before implementation. This ensures pricing consistency and prevents arbitrary price changes.

-Price tag errors: This web application will contain a section to communicate with the head office regarding any disputes related to special price tags. This will work in sync with the automated price tracking system planned above to strictly only display items on sale. For ease of use, one can filter specific prices for week periods such as "March 13-19" and only those tags will be up for possible disputes. This restriction prevents confusion regarding products that may not be on sale and displays all items with special pricing without manually checking the flyer. These disputes can range from receiving no tags, receiving tags for items that do not exist in the store and replacing damaged tags.

4. System Architecture

The system architecture will include the following components:

- Authentication & Security: A secure authentication system to allow Head Office or Admins to implement price changes.
    - User Authentication: Google OAuth for secure login via the store's Google portal.
    - Role based access control:
        - Employees can display product availability for their specific stores and manage prices.
        - Head office can supply bulk updates and manage discounts.
        - In-store managers can propose price changes, but adjustments beyond set thresholds require approval from the head office.

- Frontend/User Interface (UI): A web-based dashboard accessible via the store's Google portal, where employees can view and manage price changes.
    - A modern and natural UI with organic themes to reflect the company's tone.
    - Use React due to its lightweight, component-based structure, which fits well with small development teams and allows rapid iteration, ideal for our use case within Google portals. Head office or employees can view and update prices through a responsive UI.
    - A user-friendly wizard to guide employees through the pricing update process to minimize errors.

- Database: A cloud-based storage system to maintain historical pricing data and schedules.
    - Utilize scalable and robust tools like Amazon RDS, a relational database service on the cloud.
        - Easy scalability: Dynamically scalable according to size of data.
        - Reliability: Prevents loss of data with automated back-ups.
        - Cost efficient: The price is based on the size of data.
        - Security: has user authentication and database encryption.
    - Update product prices accurately and on time through a streamlined system.
    - Automatically applies and removes discounts without human intervention.
    - Email notification if API is down.
    - Clarifying the flow when multiple parties are involved in resolving an error.

- Integration Module: A possible API connection with the head office pricing system to receive automatic price updates.

- Make use of AWS Lambda in combination with API Gateway for smooth integration with Amazon RDS, enabling seamless and automated price uploads from the head office. AWS Lambda supports sporadic yet high-traffic tasks like mass price updates or syncing with head office systems during weekly sales changes. This dynamic scaling is suited for retail environments with fluctuating system loads.
- Leverage API connection from head office to ensure that discount updates are applied immediately using AWS Lambda.
- Automate reporting to the head office, including statistics on pricing changes, approval times, and their impact on sales.

5. Implementation Plan

Timeline:
- Week 1-2: Research and finalize system requirements, determine feasibility of head office integration.
- Week 3-4: Develop a prototype and initial UI design.
- Week 5-6: Backend development for data storage and retrieval.
- Week 7-8: Testing phase, collecting feedback from employees for usability improvements.
- Week 9: Deployment and employee training.

Resources Required:
- Google Web App development tools.
- Employee access to Google portal.
- Developer support for API integration if required.
- Testing environment and user feedback sessions.

Training Plan:
- Conduct brief training sessions for employees to introduce the system and its functionality.
- Provide an instructional guide within the Google portal for quick reference.
- Implement a feedback mechanism for employees to report issues or suggest improvements.

Data Flow Diagram:
When an Employee Logging in and Retrieving Updated Pricing Data

1. Employee Logs In
    - The employee accesses the Google Web Portal via a browser.
    - They enter their employee ID and password to authenticate.
2. Dashboard Access
    - After logging in, the employee is redirected to the dashboard.
    - A notification alert appears, indicating that new pricing updates are available.
3. Viewing Updated Prices
    - The employee clicks on the “Pricing Updates” section.
    - The system fetches the latest pricing data from the head office.

- A list of updated product prices is displayed with a timestamp.

4. Printing Reports for On-Floor Use
    - If needed, the employee clicks “Generate Printable Report” to get a PDF of the updated prices.
    - This report is printed and shared with on-floor staff.

(The diagram on the last page shows the steps outlined above visually.)

6. Analysis Of The Design

Security:
The proposed web-based pricing system must have a tight security system incase of attackers. If one were to change a price, it could apply across the whole chain, so we designed with security in mind. By using pre-existing technologies that are regularly tested, security standards are upheld with minimal effort on our part. By using Google’s OAuth we provide a secure login system with two factor authentication and a login portal to prevent phishing. React has built-in injection security as well as protection against session hijacking and brute-force attacks. AWS API gateway provides DDoS protection.

We will implement both front-end and back-end security testing to ensure comprehensive protection. On the front-end, input validation and encoding will be enforced to prevent XSS attacks, form tampering, and data injection. On the back-end, we will conduct penetration testing to detect vulnerabilities such as SQL injection, CSRF, and authentication bypass. We will follow OWASP Top 10 guidelines, and regression testing will be conducted regularly to ensure system updates do not introduce new security flaws.

There are still security risks as employees hold sensitive information creating a need for employee training. By allowing employees to make passwords we run the risk of them making a weak password, reusing passwords and even sharing them. Additionally, employees who lack cybersecurity awareness training may fall victim to social engineering. To prevent human error there must be a training program for employees.

Privacy:
By means of security and minimizing information collection, our system proves to have strong privacy protection. Ensuring privacy is important because employees must be comfortable with using the system and prices should not be publicly available. Personal privacy is not an issue as our system only collects employee names and roles. By using Google OAuth, identity management is done through their already secure system meaning there is no local storage of credentials. Price information will not be accessible as Amazon RDS provides total data encryption to protect information. Moreover, with role-based access, the information is held exclusively to the people who should have access to it.

As records are solely digital, sensitive information is limited to those who should have access. The main privacy concern is that with digital records comes reliance on the frameworks being used in the proposed design. Google and AWS are proven to be reliable in supervising data but if they were to suffer a breach then the systems access control would be compromised. Another concern, though minor, is possible misplacement of printed lists. Through training and an organization system employees must follow, this will not be a major privacy concern. A possible system stores could use is that the department manager must return all printed lists to the store manager when they are no longer needed.

Equity:
Equity is a cornerstone of the proposed pricing system for Farm Boy Lakeshore, it is simple "customer first" while ensuring fair treatment of every stakeholder, employee and manager. By maintaining open communication and stable pricing practices, the system avoids unwanted price fluctuations and mitigates previous biases which would be against the customer's interest. Our approval mechanisms ensure any verification for price variation so that all change is examined to protect consumer interests.
This structured system also promotes workers' well-being by establishing clear roles and responsibilities, thus eliminating additional stress and promoting equity within the workplace. Through live activity logs and real-time reporting, the system promotes accountability and trust between clients and the company. Through an equitable system that guarantees long term sustainability in balancing operational efficiency with ethical practice, it ultimately ensures both customers and employees reap the benefit of pricing culture, that is transparent, fair, and firmly grounded in equity, ensuring that all customers always get the right price.

Sustainability:
Sustainability was kept in mind throughout the length of the entire proposed pricing system at Farm Boy Lakeshore. Automated price updates and price tracking are fully available online in contrast to traditional methods that require a manual process. From a waste standpoint, this system greatly reduces the amount of papers regularly printed on the day of price changes at the store to eliminate duplicate tags or price checking items. The reduction of paper usage in this core system will then contribute to lower waste production at the store entirely.

From a technological perspective, by integrating this pricing system into the already existing Google based web portal the store has ensured all proposed features are smoothly embedded without the development and maintenance of a separate website. This would eliminate the usage of additional API connections from the store alongside the already proposed API connections from head office. By doing this, system complexity is simplified and avoids the risk of additional emissions. Our backend system of receiving price updates through AWS Lambda and API Gateway is made to support automated pricing updates without servers running continuously. AWS Lambda's event driven features support these high traffic tasks such as weekly or monthly price changes without computing these large changes unnecessarily during idle periods. This simple system is cost effective and reduces emissions by reducing computing when there are no active price changes and systems are not in use.

Ethical consideration:
Implementing the proposed pricing system at Farm Boy Lakeshore requires addressing specific ethical considerations to ensure responsible and fair practices. Ethical responsibility centers around transparent communication of pricing to avoid misleading customers. The system must clearly display accurate and timely price information, quickly resolving any discrepancies that arise to maintain customer trust.

Fairness is critical; automated pricing systems must avoid unintended biases or unjustified price fluctuations. Ethical oversight demands that price adjustments undergo appropriate scrutiny and authorization processes to protect against manipulation and ensure equitable treatment of all customers.

Employee roles within the system must also be ethically managed, ensuring the technology is supportive rather than punitive. Employees should not face adverse impacts from automated systems such as increased pressure or unfair accountability for systematic errors. Ethical practice requires clear guidelines that define roles and responsibilities, fostering a supportive rather than adversarial workplace environment.

Moreover, ethical responsibility includes a commitment to openness regarding system operations, with transparency about how decisions and changes are made. This approach builds trust among stakeholders and supports an organizational culture committed to integrity.

Ethical considerations must ensure accountability by maintaining accurate and comprehensive audit trails of all price adjustments. Such transparency is critical to verifying compliance with ethical standards, reinforcing the system's integrity, and maintaining trust with all stakeholders.

Societal Impact:
The proposed pricing management system presents a range of societal benefits, but it also introduces challenges that must be carefully managed. On the positive side, improved pricing accuracy enhances the customer experience by fostering trust and reducing instances of price-related confusion or disputes. This transparency strengthens the relationship between Farm Boy and its customer base. Additionally, automation reduces the manual burden on employees, allowing them to focus on higher-value tasks, which can improve job satisfaction and operational efficiency. However, this shift may also lead to concerns about job displacement or a perceived devaluation of employees' roles if not accompanied by adequate training and reassurances about job security. The centralized control of pricing decisions increases consistency across locations, but it may also limit the autonomy of store managers who possess valuable on-the-ground insights. If their input is overlooked, it could lead to frustration or a disconnect between head office decisions and local customer needs. To mitigate this, implementing a structured feedback loop that empowers staff to influence future system iterations is essential. Overall, while the system has the potential to improve fairness, efficiency, and trust, these gains must be balanced with deliberate efforts to maintain employee agency and ensure an inclusive transition.

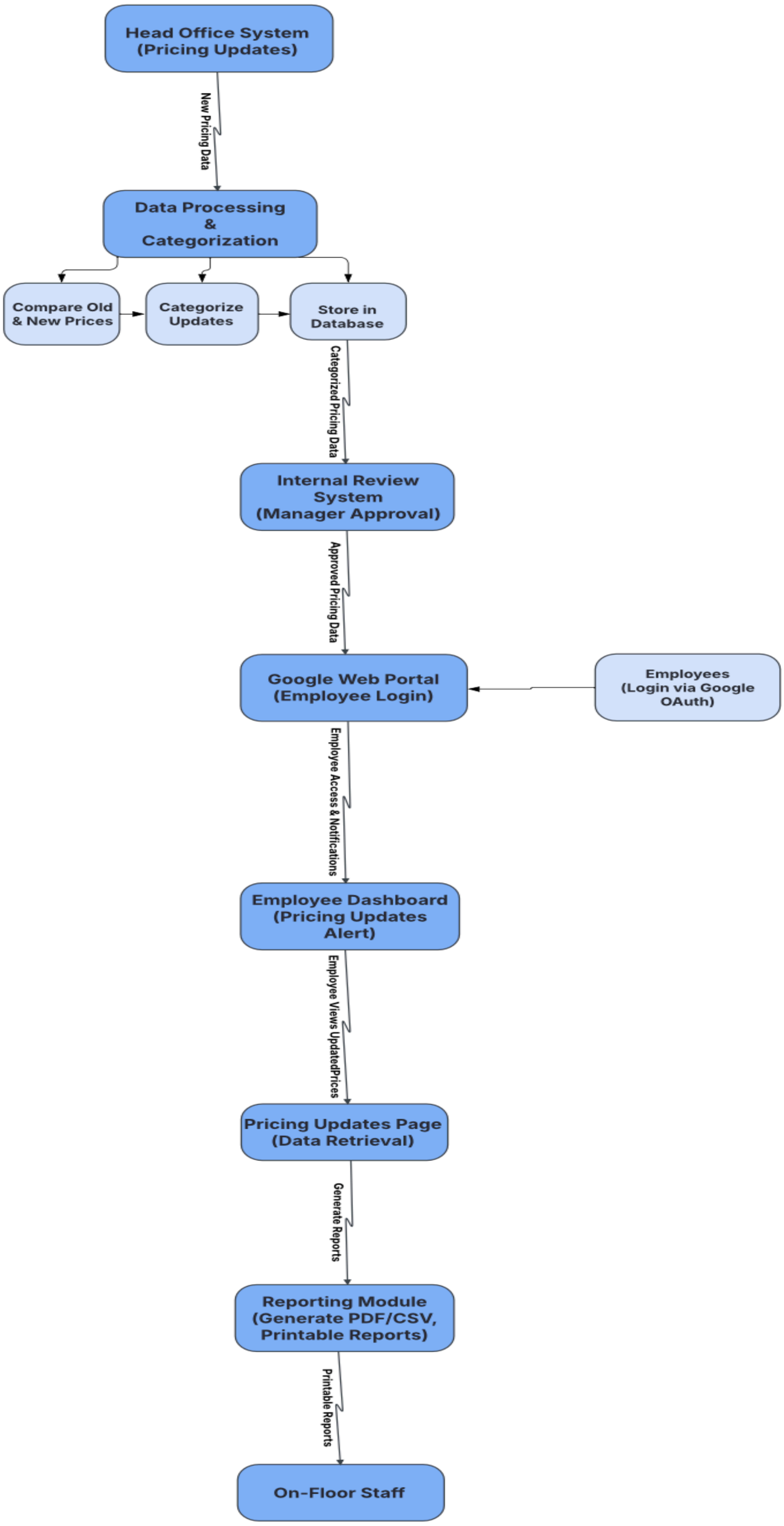
7. Conclusion

The proposed digital pricing system offers a promising solution to the operational and communication challenges faced by Farm Boy Lakeshore. By automating price tracking, streamlining employee workflows, and integrating with existing Google-based systems, the solution can significantly reduce pricing errors, improve consistency across departments, and enhance overall efficiency. Key features such as role-based access, printable reports, and real-time updates from the head office promote accountability and transparency in price management.

However, successful implementation will require careful coordination, particularly during employee training and system testing. While there may be initial resistance to adopting a new system, thoughtful deployment and user adoption strategies will be crucial. Continuous evaluation and support will ensure the long-term success of the system, as it evolves to meet the operational needs of the store.

Security and privacy, although well-considered in the design, rely heavily on proper employee training and adherence to best practices. While integration with the head office system offers a powerful feature, it does introduce technical complexity. Nevertheless, the system has been designed to minimize these risks through robust security measures and scalable infrastructure.

In conclusion, the proposed pricing system provides Farm Boy Lakeshore with a solid foundation for addressing current inefficiencies while enabling future growth. By ensuring pricing consistency, reducing manual tasks, and improving transparency, this system will help Farm Boy deliver a better customer experience, while also supporting operational efficiency. With proper implementation and support, this system will position Farm Boy for long-term success and sustainability.


Head Office System (Pricing Updates)
New Pricing Data
Data Processing & Categorization
Compare Old & New Prices
Categorize Updates
Store in Database
Categorized Pricing Data
Internal Review System (Manager Approval)
Approved Pricing Data
Google Web Portal (Employee Login)
Employees (Login via Google OAuth)
Employee Access & Notifications
Employee Dashboard (Pricing Updates Alert)
Employee Views UpdatedPrices
Pricing Updates Page (Data Retrieval)
Generate Reports
Reporting Module (Generate PDF/CSV, Printable Reports)
Printable Reports
On-Floor Staff